

An Integrated ROS 2 Stack for Humanoid Locomotion, Autonomous Navigation, and Conversational Interaction on the Unitree G1 EDU

M. A. Abdul Rafi¹

¹Department of Computer Science & Engineering
University of Moratuwa, Sri Lanka
Email: abdulr.23@cse.mrt.ac.lk

Abstract

Deploying a humanoid robot as a passenger-facing service agent requires the integration of three normally separate research areas: learned bipedal locomotion, LiDAR-based indoor mapping and navigation, and natural, voice-driven human-robot interaction. This paper presents the design and implementation of a ROS 2 Jazzy software stack for the Unitree G1 EDU humanoid, developed as a university capstone project in collaboration with a ride-hailing platform to prototype an airport mobility concierge. The system comprises three layers: an ONNX-based reinforcement-learning locomotion policy driving a state-machine teleoperation and control node validated in MuJoCo simulation; a mapping and navigation layer combining live voxelized 3D LiDAR mapping with SLAM Toolbox and Nav2 for point-cloud filtering, localization, and path planning; and a Retrieval-Augmented Generation (RAG) conversational agent that transcribes speech with Whisper, retrieves grounded answers from a FAISS-indexed knowledge base using a LangChain agent, and replies through synthesized speech, enabling multilingual, privacy-preserving passenger interaction. A React-based web gateway exposes teleoperation, mapping, and navigation-goal controls to a browser client over the robot's onboard network. Simulated evaluation shows that the locomotion policy produces stable walking gaits under keyboard teleoperation, the mapping and navigation layer builds consistent voxel maps and executes Nav2 goal-directed traversal, and the conversational agent reliably answers in-domain queries across multiple languages while discarding passenger-identifying data at the end of each session. The results demonstrate that a single ROS 2 architecture can coherently span learned control, autonomous navigation, and language-grounded interaction for a real-world humanoid service application.

Index Terms—Humanoid Robotics, ROS 2, Reinforcement Learning, SLAM, Nav2, Retrieval-Augmented Generation, Human-Robot Interaction

I. INTRODUCTION

Humanoid robots are increasingly proposed as general-purpose service agents capable of operating in human-centric spaces such as airports, retail environments, and transit hubs, where a bipedal form factor allows navigation through infrastructure designed for people. Realizing such a role requires far more than a single capability: the robot must walk reliably under dynamic control, build and use a map of an unfamiliar indoor environment to reach a destination, and communicate naturally with the people it serves, all coordinated through a common software architecture.

Most published humanoid research addresses these problems in isolation, learned locomotion policies are typically demonstrated on treadmills or in controlled gait-tracking benchmarks, SLAM and navigation stacks are most often validated on wheeled platforms, and conversational agents are studied largely as text or voice interfaces detached from an embodied robot. This paper instead describes a single, integrated ROS 2 system built for the Unitree G1 EDU humanoid that couples these three layers into one operational

stack, applied to a concrete use case, a passenger-facing mobility concierge developed with a ride-hailing partner.

The main contributions of this paper are: (1) a layered ROS 2 architecture that connects an ONNX reinforcement-learning locomotion policy, a SLAM Toolbox/Nav2 mapping and navigation pipeline, and a Whisper/RAG conversational agent under a common communication backbone spanning MQTT, DDS, and WebRTC; (2) a browser-based control gateway that exposes teleoperation, live mapping, and navigation goals to a remote operator without native ROS 2 tooling; and (3) a privacy-conscious, multilingual voice concierge pipeline that grounds its responses in an approved knowledge base and discards passenger data at the end of each session.

II. LITERATURE REVIEW

ROS 2 was designed to meet the real-time, multi-robot, and safety-critical requirements that limited the original ROS in production robotic deployments, and provides the middleware backbone, DDS-based communication, and lifecycle management used throughout this system [1]. For learned locomotion, Rudin *et al.* showed that massively parallel

GPU-based reinforcement learning can train stable legged-locomotion policies for a quadrupedal robot in minutes rather than days, using a curriculum-driven training regime later adapted across many legged and humanoid platforms [2]; the present system deploys a policy of this general class, trained and simulated with MuJoCo, a physics engine purpose-built for model-based control and contact-rich articulated-body simulation that has become a standard tool for both classical control and reinforcement-learning research [3].

For mapping and navigation, SLAM Toolbox provides pose-graph-based 2D/3D simultaneous localization and mapping with support for continued mapping and optimization-based localization, and is the currently recommended SLAM package for ROS 2 [4], while Navigation2 (Nav2) reimplements the classical ROS Navigation stack using behaviour trees and lifecycle-managed nodes suited to the real-time requirements of ROS 2, and has been validated in long-duration deployments alongside pedestrians [5].

For the conversational layer, Radford *et al.* introduced Whisper, a speech-recognition model trained on 680,000 hours of weakly supervised multilingual and multitask audio that generalizes to new domains without task-specific fine-tuning, which this system uses for on-device speech transcription [6]. To ground the conversational agent’s responses in an approved knowledge base rather than relying purely on a language model’s parametric memory, this work adopts the Retrieval-Augmented Generation (RAG) paradigm of Lewis *et al.*, which couples a pre-trained generator with a non-parametric retriever over an external document index to improve factuality and provenance [7]. The retrieval index itself is built with FAISS, a library for efficient similarity search over large, high-dimensional vector collections that underlies the dense retrieval step in many modern RAG systems [8].

III. SYSTEM ARCHITECTURE AND METHODS

A. Hardware and Communication Architecture

The Unitree G1 EDU exposes two onboard computers alongside a cloud service and a manufacturer mobile application, communicating over a combination of MQTT, HTTP, WebRTC, BLE, and DDS. The ROS 2 Jazzy workspace developed in this work runs across these onboard computers and interfaces with the robot’s native control layer, external developer tooling, and a browser-based operator interface, all coordinated through this heterogeneous transport stack rather than a single uniform bus.

B. Locomotion Layer

Bipedal walking is driven by an ONNX-exported reinforcement-learning policy of the class described by Rudin *et al.* [2], wrapped in a ROS 2 state-machine node that consumes velocity commands on `/cmd_vel` and outputs joint-level actuation targets. A keyboard teleoperation node maps forward/backward, turning, strafing, speed adjustment, and stop commands to this velocity interface for manual control and testing. Prior to any hardware deployment, the full locomotion and control loop is validated against a MuJoCo-based simulator [3], bridged into ROS 2 so that the

simulated G1 exposes the same topic interface as the physical robot, allowing locomotion, mapping, and navigation code to be developed and tested without hardware risk.

C. Mapping and Navigation Layer

A dedicated mapping node builds a live voxelized 3D representation of the environment from LiDAR point clouds, visualized in RViz for operator inspection. This map feeds a navigation layer that integrates SLAM Toolbox for pose-graph localization and mapping [4] with Nav2 for costmap-based path planning and execution [5], together with point-cloud filtering and a command-arbitration node that reconciles autonomous navigation commands with manual overrides on a shared `/cmd_vel_safe` topic. A browser-based control gateway, built with React and connected through the robot’s onboard network, allows an operator to drive the robot, monitor the live map, and issue navigation goals without installing native ROS 2 client software.

D. Conversational Concierge Pipeline

The flagship application layer turns the G1 into a passenger-facing concierge that explains a partner ride-hailing service and assists with its mobile-app installation. Speech is detected using WebRTC-based voice activity detection and transcribed locally with Whisper [6]. The transcribed query is passed to a LangChain-orchestrated agent that retrieves relevant passages from a FAISS-indexed knowledge base [8] covering the partner service, its installation steps, and local location data, embedded through a large language model, following the retrieval-augmented generation pattern of Lewis *et al.* [7] to keep responses grounded in approved sources. Generated responses are synthesized to speech and played through the robot’s onboard speaker. The pipeline supports onboarding and mid-conversation language switching across eleven languages, and intermediate pipeline state, transcriptions, agent responses, and session events, is published as ROS 2 topics for monitoring. Passenger name and session data are cleared immediately at session end, so no personally identifying information persists between interactions.

IV. RESULTS AND DISCUSSION

A. Performance Analysis

The system was evaluated primarily in the MuJoCo-based simulation environment, with each layer assessed independently before end-to-end integration.

- **Locomotion:** the ONNX policy, driven through the keyboard teleoperation interface, produced stable forward, backward, turning, and strafing gaits in simulation across the tested command range.
- **Mapping and navigation:** the mapping node produced consistent, incrementally refined voxel maps of the simulated environment, and the SLAM Toolbox/Nav2 pipeline successfully localized the robot and executed goal-directed traversal to operator-specified navigation targets.
- **Conversational agent:** the Whisper/RAG pipeline correctly transcribed and answered in-domain spoken queries

about the partner service across the supported languages, and consistently discarded session-specific passenger data once a conversation concluded.

B. Challenges and Limitations

Several integration challenges arose from combining these three layers. The heterogeneous transport stack, spanning MQTT, DDS, WebRTC, and BLE across the cloud service, onboard computers, and manufacturer app, required careful arbitration to avoid command conflicts between autonomous navigation and manual teleoperation, addressed through the dedicated command-arbitration node publishing to `/cmd_vel_safe`. Because development and validation were carried out primarily in simulation using MuJoCo ground-truth odometry, deployment on physical hardware requires replacing this with real LiDAR-inertial odometry or SLAM, verifying the full `map→odom→base_footprint→pelvis→mid360_link` transform chain, and re-tuning footprint, obstacle-height bands, and stop distances, which are conservative starting points rather than calibrated safety limits. The current navigation stack also assumes flat indoor floors and does not yet handle drop-offs, stairs, or slopes, and the conversational agent’s Phase 1 scope remains informational, ride booking through the concierge is not yet implemented.

V. CONCLUSION AND FUTURE WORK

This work presented an integrated ROS 2 Jazzy stack for the Unitree G1 EDU humanoid that couples an ONNX reinforcement-learning locomotion policy, a SLAM Toolbox and Nav2 mapping and navigation pipeline, and a Whisper/RAG multilingual conversational concierge, connected through a heterogeneous communication architecture and exposed through a browser-based control gateway. Simulated evaluation shows that the three layers function both independently and together, producing stable locomotion, consistent mapping and goal-directed navigation, and grounded, privacy-conscious conversational responses, demonstrating the feasibility of a single coherent software architecture for humanoid service robotics.

Future work will focus on transferring the locomotion and navigation layers to physical hardware with real LiDAR-inertial odometry, extending the navigation stack to handle non-flat terrain such as stairs and slopes, expanding the conversational agent’s scope from informational assistance to full ride-booking interaction, and conducting user studies to evaluate the concierge’s effectiveness with real passengers in an airport setting.

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